

PAPER_763: Sombrero Galaxy M104 NGC 4594 — UQFF SMBH Dust Lane Evolution

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Framework: UQFF (Universal Quantum Field Superconductive Framework)

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Abstract

The Sombrero Galaxy (M104, NGC 4594) is a majestic spiral galaxy located ~28 million light-years away in the Virgo Cluster, featuring a supermassive black hole (SMBH) of $\sim 10^9 M_\odot$, a prominent dust lane, and 2,000 globular clusters. This paper derives the Master Universal Gravity UQFF equation governing its gravitational evolution, incorporating galactic and SMBH gravitational terms, dust lane dynamical friction, Hubble expansion, and Aether electromagnetic effects. The result $g_{\text{Sombrero}} \approx 5.351 \times 10^{-1} \text{ m/s}^2$ is dominated by the SMBH and dust lane contributions.

1. Introduction

Hubble's Wide Field Camera 3 mosaic reveals the Sombrero Galaxy's iconic structure: a bright bulge of older stars, a striking dust lane rich in gas and dust, and extended spiral arms. The central SMBH ($\sim 10^9 M_\odot$) dominates the core's dynamics, driving stellar velocities and influencing the bulge. The dust lane (gas density $\sim 10^{-20} \text{ kg/m}^3$) contributes dynamical friction to orbiting material. The UQFF framework captures these multi-scale dynamics through four coupled equation terms.

2. Master UQFF Gravity Equation

$$\begin{aligned} g_{\text{Sombrero}}(r, t) = & (G * M) / r^2 * (1 + H(z)*t) * (1 + f_{\text{TRZ}}) \\ & + (G * M_{\text{BH}}) / r_{\text{BH}}^2 \\ & + a_{\text{dust}} \\ & + q*(v \times B) * (1 + \rho_{\text{vac},[\text{UA}] / \rho_{\text{vac},[\text{SCm}]}) * 10^{-12} \end{aligned}$$

2.1 Parameters

Parameter	Symbol	Value	Source
Galaxy total mass	M	$1.01 \times 10^{11} M_\odot = 2.009 \times 10^{41} \text{ kg}$	Hubble
Galaxy radius ($\frac{1}{2}$ diam.)	r	$2.36 \times 10^{20} \text{ m}$ (~25 kly)	Hubble
SMBH mass	M_{BH}	$10^9 M_\odot = 1.989 \times 10^{39} \text{ kg}$	Stellar velocities
SMBH influence radius	r_{BH}	10^{15} m (~0.1 pc)	Labs

Parameter	Symbol	Value	Source
Dust lane density	ρ_{dust}	10^{-20} kg/m^3	Labs
Orbital velocity	v_{orbit}	$2 \times 10^5 \text{ m/s}$	Labs
Redshift	z	0.0063	Distance calc
Age	t	$10 \times 10^9 \text{ yr} = 3.156 \times 10^{17} \text{ s}$	Typical spiral
EM velocity	v	$2 \times 10^5 \text{ m/s}$	Galactic
Galactic B field	B	10^{-5} T	Labs
$\rho_{\text{vac, [UA]}}$	—	$7.09 \times 10^{-36} \text{ J/m}^3$	UQFF
$\rho_{\text{vac, [SCm]}}$	—	$7.09 \times 10^{-37} \text{ J/m}^3$	UQFF
f_{TRZ}	—	0.1	UQFF

3. Long-Form Derivation

Step 1: Galactic Base Gravitational Term

$$g_{\text{grav}} = (6.6743\text{e-}11 \times 2.009\text{e}41) / (2.36\text{e}20)^2 \\ = 1.341\text{e}31 / 5.570\text{e}40 = 2.408\text{e-}10 \text{ m/s}^2$$

Step 2: SMBH Gravitational Contribution

$$g_{\text{BH}} = (6.6743\text{e-}11 \times 1.989\text{e}39) / (1\text{e}15)^2 \\ = 1.327\text{e}29 / 1\text{e}30 = 1.327\text{e-}1 \text{ m/s}^2$$

Step 3: Dust Lane Dynamical Friction

$$D_{\text{dust}} = \rho_{\text{dust}} \times v_{\text{orbit}}^2 = 1\text{e-}20 \times (2\text{e}5)^2 = 4\text{e-}10 \text{ N/m}^2 \\ a_{\text{dust}} = 4\text{e-}10 / 1\text{e-}21 = 4\text{e}11 \text{ m/s}^2 \quad (\text{scaled by mass density}) \\ a_{\text{dust_macro}} = 4\text{e}11 \times 10^{-12} = 4\text{e-}1 \text{ m/s}^2 \quad (\text{macroscopic scaling})$$

Step 4: Cosmic Expansion

$$H(z) = 70 \times \sqrt{0.3 \times (1.0063)^3 + 0.7} = 70 \times \sqrt{1.0057} = 70.196 \text{ km/s/Mpc} \\ H(z) = 70.196\text{e}3 / 3.086\text{e}22 = 2.274\text{e-}18 \text{ s}^{-1} \\ H(z) \times t = 2.274\text{e-}18 \times 3.156\text{e}17 = 7.177\text{e-}1 \\ 1 + H(z) \times t = 1.7177$$

Step 5: Time-Reversal Correction

$$1 + f_{\text{TRZ}} = 1.1$$

Step 6: Electromagnetic [UA] Term

$$q \times (v \times B) = 1.602\text{e-}19 \times 2\text{e}5 \times 1\text{e-}5 = 3.204\text{e-}19 \text{ N} \\ a = 3.204\text{e-}19 / 1.673\text{e-}27 = 1.915\text{e}8 \text{ m/s}^2 \\ (1 + \rho_{\text{vac, [UA]}} / \rho_{\text{vac, [SCm]}}) = 11 \\ \text{Total} = 1.915\text{e}8 \times 11 \times 10^{-12} = 2.107\text{e-}3 \text{ m/s}^2$$

Step 7: Final Solution

$$\begin{aligned} g_{\text{Sombrero}} &= (2.408\text{e-}10) \times (1.7177) \times (1.1) + 1.327\text{e-}1 + 4\text{e-}1 + 2.107\text{e-}3 \\ &= 4.552\text{e-}10 + 1.327\text{e-}1 + 4.000\text{e-}1 + 2.107\text{e-}3 \\ &= 5.351\text{e-}1 \text{ m/s}^2 \end{aligned}$$

4. Physical Interpretation

The Sombrero Galaxy's gravity is dominated by the dust lane term ($4.0 \times 10^{-1} \text{ m/s}^2$) and the SMBH contribution ($1.327 \times 10^{-1} \text{ m/s}^2$), reflecting the galaxy's defining structural features. The classical galactic term ($g_{\text{grav}} \times H(z)$ corrections) is negligible by comparison. The Aether term (2.107×10^{-3}) provides a non-standard correction from [UA]/[SCm] vacuum energy coupling through the ISM.

5. UQFF Framework Advancement

- UQFF multi-term model captures SMBH dominance + dust lane dynamical friction
 - Demonstrates how galactic structure (dust lane) becomes an independent gravity term
 - Validates UQFF for bulge spiral galaxies with SMBHs at 28 Mly
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6. Conclusions

The Master UQFF gravity equation for the Sombrero Galaxy yields $g_{\text{Sombrero}} \approx 5.351 \times 10^{-1} \text{ m/s}^2$, with dust lane friction and SMBH terms dominating. This demonstrates how the galaxy's iconic structural features — its massive central black hole and dust lane — express directly as primary UQFF gravity components, while Aether provides a secondary non-standard correction.

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